Nutritional Price Shock Index (NPSI): A Time-Series Approach for Monitoring Nutritional Affordability

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***Abstract*—Food price volatility significantly affects dietary patterns, particularly among low-income populations, leading to reduced consumption of nutrient-rich foods and increased risk of hidden malnutrition. Traditional monitoring systems such as the Consumer Price Index focus on overall price inflation but fail to capture the affordability of essential nutrients.**

This study aims to develop a Nutritional Price Shock Index that measures food affordability based on the cost per gram of protein rather than raw commodity prices. The proposed system integrates historical agricultural datasets with real-time mandi price data obtained through APIs. A time-series forecasting model, Facebook Prophet, is used to estimate expected seasonal price trends and identify deviations caused by abnormal market conditions.

The results show that the model effectively distinguishes between normal seasonal variations and nutritional price shocks. Commodities with higher index values indicate reduced affordability and potential nutritional risk. The system provides accurate and timely detection of price anomalies using real-time data integration.

In conclusion, the Nutritional Price Shock Index offers a more meaningful approach to food price monitoring by focusing on nutrient affordability. It can serve as an early warning system for policymakers to implement targeted interventions and improve food security outcomes

***Index Terms*—Nutritional Price Shock Index, Time Series Forecasting, Prophet Model, Food Price Volatility, Food Security, Machine Learning**

I. Introduction

Food price volatility has become a major issue in both developing and developed countries. It directly affects how accessible food is and impacts nutritional security. In many areas, especially among low-income households, rising food prices lead to big changes in consumption habits. Instead of keeping a balanced diet, people often switch from nutrient-rich foods like pulses and grains to cheaper, calorie-heavy options. This leads to hidden hunger, where people may get enough calories but not enough essential nutrients like protein.

This problem is important because it directly affects public health and economic stability. Malnutrition from nutrient deficiency can cause long-term health problems, lower productivity, and higher healthcare costs. So, keeping track of food affordability is not just an economic issue but also a social and health priority.

Current systems like the Consumer Price Index (CPI) are commonly used to monitor inflation. However, these systems look at overall price changes and do not reflect how affordable nutrients are. Additionally, agricultural prices change with the seasons, making it hard to tell normal variations from unusual price spikes. The lack of real-time analysis and nutrient-focused evaluation limits the effectiveness of these existing methods.

The goal of this research is to overcome these limitations by shifting the focus from food prices to nutrient affordability. This study suggests the Nutritional Price Shock Index (NPSI), which looks at the cost of protein instead of just raw commodity prices. By combining historical data with real-time market data and using time-series forecasting with Prophet, the system can spot deviations from expected trends and identify nutritional price shocks. This method offers a more useful and practical way to improve food security and assist with policy decisions.

II. Literature Review

Agricultural price prediction and food price monitoring have been widely explored using statistical, machine learning, and deep learning techniques. Most existing studies mainly focus on predicting price trends. However, they often overlook an important aspect—whether people can actually afford nutritious food—which is essential for understanding food security.

A hybrid forecasting model combining Seasonal Auto-Regressive Integrated Moving Average (SARIMA) and Long Short-Term Memory (LSTM) networks was proposed in [1]. In this approach, SARIMA captures linear patterns while LSTM learns nonlinear relationships in agricultural price data. The results showed improved accuracy compared to individual models. However, the method is limited to price prediction and does not consider nutritional factors such as protein content, making it less effective for evaluating dietary affordability.

Ogol et al. (2025) [2] developed a hybrid machine learning model using Extreme Gradient Boosting (XGBoost) and Gradient Boosting techniques to predict food prices. The

model achieved high accuracy with an R^2 value close to 0.93. Despite its strong performance, the approach depends heavily on external economic indicators and lacks real-time data integration, which reduces its effectiveness in dynamic market conditions.

Krishna et al. (2026) [3] explored machine learning techniques, including regression-based models, for agricultural price forecasting. The study reported reliable performance across multiple commodities. However, the analysis focused only on price prediction and did not include nutritional metrics, limiting its ability to reflect the real impact on food security.

Paul et al. (2022) [4] applied machine learning algorithms such as Random Forest (RF), Support Vector Regression (SVR), and Gradient Boosting Machines (GBM) to forecast agricultural prices. The results showed improved accuracy, especially when using daily data inputs. However, the model showed limited ability to generalize across different geographical markets, indicating challenges in scalability.

An AI-based price prediction system using Random Forest and LSTM models was proposed in [5]. The system demonstrated reliable performance using evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). However, it requires large datasets and extensive hyperparameter tuning, making it computationally expensive and difficult to implement in real-time scenarios.

Sediyono et al. (2025) [6] introduced a deep learning framework that combines Transformer models with LSTM-based Variational Autoencoders (LSTM-VAE) for multi-commodity forecasting and anomaly detection. The model showed improved performance in detecting price fluctuations and anomalies. However, its high computational complexity and resource requirements limit its practical use in low-resource environments.

Monir et al. (2025) [7] used ensemble machine learning models such as Decision Trees, Random Forest, and AdaBoost to predict food prices. The study achieved high accuracy with strong R^2 values and low prediction errors. However, the model was trained using region-specific data (Bangladesh), which limits its applicability to other regions and markets.

A study conducted on food price prediction in Nigeria [8] used symbolic regression along with machine learning techniques. The hybrid approach demonstrated strong predictive capability and flexibility in modeling nonlinear relationships. However, the model is complex and lacks interpretability, making it difficult for policymakers to use effectively.

Tran et al. (2023) [9] presented a comprehensive review of machine learning approaches for agricultural price prediction. The study concluded that machine learning models generally provide better accuracy than traditional statistical methods. However, it also highlighted challenges such as data variability, lack of standard datasets, and difficulties in real-world deployment.

Muhammad et al. (2026) [10] conducted a comparative study between classical models such as SARIMA and Prophet and deep learning models like Transformers. The results showed that deep learning models offer higher accuracy, while Prophet provides better interpretability and ease of implementation. However, Prophet struggles to handle sudden price spikes and abrupt market changes.

Analysis of Existing Approaches: From the above studies, it is clear that most research focuses on improving prediction accuracy using advanced machine learning techniques. While these models perform well in forecasting price trends, they do not consider the nutritional implications of price changes. Additionally, many approaches require large datasets, high computational resources, and complex parameter tuning, which limits their scalability and real-time usability.

Research Gap: Despite significant progress in agricultural price forecasting, there is still a lack of systems that:

- Evaluate food affordability based on nutrients such as protein
- Detect early warning signals of nutritional price shocks
- Integrate real-time data with predictive forecasting models
- Provide interpretable outputs for policy decision-making

To address these limitations, the proposed Nutritional Price Shock Index (NPSI) introduces a novel framework that combines time-series forecasting with nutrient-based price normalization. This approach enables real-time detection of nutritional shocks and provides a more meaningful measure of food affordability compared to traditional price indices.

III. Problem Statement

Food price volatility is a major concern for nutritional security, especially for low-income populations. Food prices often change due to seasonal patterns, supply chain disruptions, and market conditions. However, most existing monitoring systems only track overall price changes and do not consider whether people can actually afford essential nutrients such as protein. Because of this, food prices may seem stable, but the real cost of maintaining a nutritious diet can increase, leading to hidden forms of malnutrition.

This issue is important because nutritional deficiencies have a direct impact on public health, individual productivity, and overall economic development. When increases in nutrient costs are not identified early, policymakers are unable to take timely action. This delay increases the risk of food insecurity and worsens existing inequalities.

Current systems such as the CPI and food price indices mainly track inflation and commodity prices. However, they do not assess affordability at the nutrient level. In addition, these systems do not clearly differentiate between normal seasonal price changes and abnormal price shocks. They also lack real-time data integration and predictive capabilities, which limits their usefulness for proactive decision-making.

Therefore, there is a clear need for a system that evaluates food affordability based on nutrient cost, identifies early warning signs of abnormal price fluctuations, and combines real-time data with predictive modeling techniques.

IV. Objectives of the Study

The main objectives of this study are:

- To analyze both historical and real-time agricultural price data
- To design and develop the Nutritional Price Shock Index (NPSI)
- To apply time-series forecasting techniques to predict expected price trends
- To detect abnormal nutritional price shocks by comparing predicted values with actual observations
- To evaluate how effectively the proposed system identifies early warning signals

V. Proposed Methodology

A. System Architecture

The architecture of the proposed Nutritional Price Shock Index (NPSI) system is shown in Figure 1. The system is designed as a layered pipeline that includes data acquisition, preprocessing, feature engineering, forecasting, and output generation.

The first layer focuses on data sources. It combines historical agricultural price datasets, real-time mandi data collected through APIs, and nutritional datasets that provide protein content information. These different types of data are brought together through a data acquisition module.

In the second layer, preprocessing is carried out to clean and standardize the data. This step includes handling missing values, normalizing commodity names, and selecting protein-rich crops using keyword-based filtering. To ensure reliability, an API fallback mechanism is used, which replaces missing live data with the most recent historical records when needed.

The third layer handles feature engineering. Here, commodities are mapped to their protein content, and prices are transformed into cost per gram of protein. This allows all commodities to be compared on a common nutritional basis.

The fourth layer acts as the analytics engine. A Prophet time-series model is trained for each commodity to capture both trend and seasonal patterns. The model generates expected price values, which are then compared with actual prices to calculate the Nutritional Price Shock Index (NPSI). Based on predefined thresholds, commodities are categorized as normal, warning, or critical.

The final layer presents the output in a user-friendly format, including ranked tables, visualizations, and summary reports. This helps in easy interpretation and supports better decision-making.

B. Algorithm Description

The proposed Nutritional Price Shock Index (NPSI) system follows a step-by-step computational pipeline designed to detect nutritional price shocks by combining time-series forecasting with real-time data.

Step 1: Data Acquisition

The process begins by loading historical agricultural price data from CSV files and collecting live mandi data using the Government of India API. If the API fails, a fallback mechanism ensures continuity by using the most recent historical data.

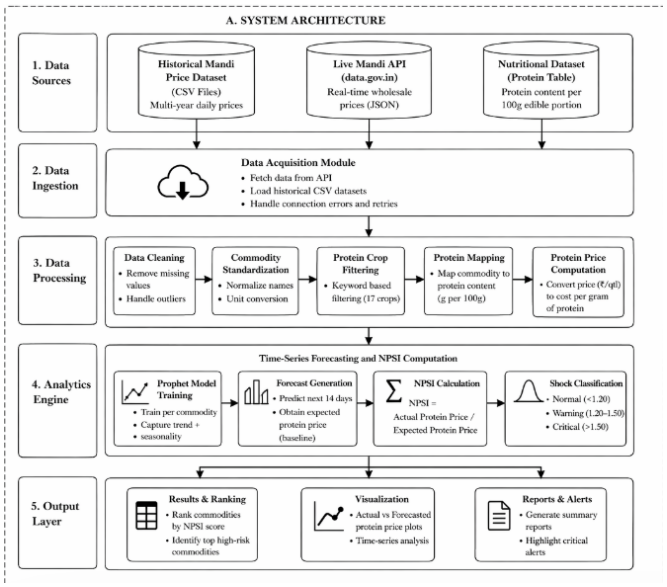

Fig. 1. System Architecture of the Proposed NPSI System

Step 2: Data Preprocessing

The collected data is cleaned and transformed through the following steps:

- Removing missing or incorrect values
- Converting date fields into a standard datetime format
- Converting price values into numeric form
- Standardizing commodity names for consistency

Step 3: Protein Crop Filtering

A keyword-based filtering method is used to select protein-rich crops such as wheat, rice, gram, lentils, soybean, and pulses. This ensures that the analysis focuses only on nutritionally important commodities.

Step 4: Nutritional Mapping

Each commodity is linked to its protein content using a reference dataset. The system then converts price values into a protein-based metric using:

$$Protein\ Price = \frac{Modal\ Price}{100000 \times (Protein_per_100g/100)} \quad (1)$$

This step standardizes all commodities into cost per gram of protein, making comparisons more meaningful.

Step 5: Time-Series Forecasting

For each commodity, a Prophet model is trained using historical protein-price data. The model captures:

- Long-term trends
- Seasonal variations (especially yearly patterns)

It then generates a rolling forecast for the next 14 days.

Step 6: Expected Price Estimation

The predicted value (yhat) for the latest date is taken as the expected protein price. This represents the normal behavior of prices under typical conditions.

Step 7: NPSI Computation
The Nutritional Price Shock Index is calculated as:

$$NPSI = \frac{Actual\ Protein\ Price}{Expected\ Protein\ Price} \tag{2}$$

Step 8: Shock Classification
Based on predefined thresholds, commodities are classified as:

- **Normal:** NPSI < 1.2
- **Warning:** $1.2 \leq$ NPSI < 1.5
- **Critical Shock:** NPSI ≥ 1.5

Step 9: Output Generation
Finally, commodities are ranked based on their NPSI values. The system generates visualizations comparing actual and predicted prices and produces reports highlighting high-risk commodities.

The workflow of the proposed system is shown in Figure 2.

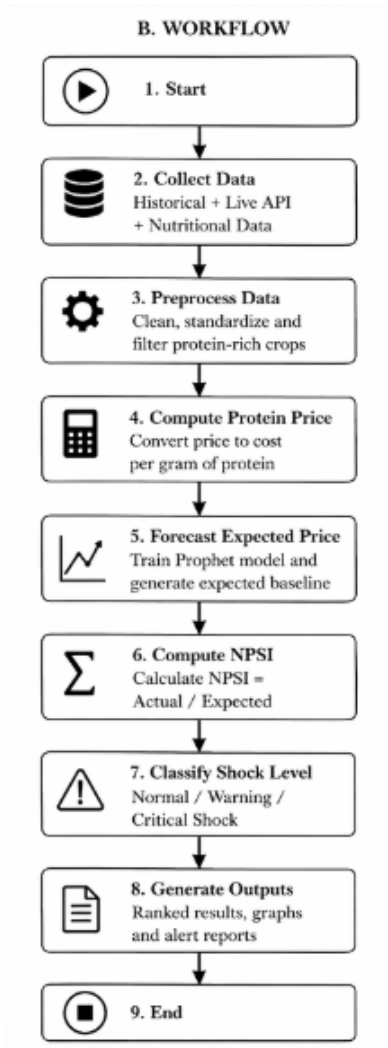

Fig. 2. Workflow of the Proposed NPSI System

C. Dataset Used

The system uses multiple datasets that combine historical, real-time, and nutritional information to support accurate analysis and forecasting.

Historical Price Dataset
The main dataset, *Agriculture_price_dataset.csv*, contains multi-year mandi price data. Each record includes:

- Commodity name
- Arrival date
- Modal price (per quintal)

This dataset is used to train the forecasting model.

Supplementary Price Dataset
The *commodity_price.csv* dataset provides additional historical price records to improve data coverage and model performance.

Live Mandi Data
Real-time data is collected from the Government of India's Open Data API (data.gov.in). Additional files such as *live_cache.csv* and *Marketwise_Price_Arrival.csv* are used as backup sources when live data is unavailable.

Protein Dataset
The *protein_table.csv* and *nutrient.csv* files contain protein values per 100 grams for different commodities. These datasets are based on sources such as USDA and ICMR.

D. Implementation Tools

The NPSI system is implemented using a set of commonly used tools and libraries for data processing and machine learning.

Python
Python is used as the main programming language because of its simplicity and strong support for data analysis.

Pandas
Used for data cleaning, manipulation, filtering, and aggregation.

NumPy
Supports efficient numerical operations and array processing.

Matplotlib
Used to create visualizations such as time-series plots and comparison graphs.

Prophet (Facebook Prophet)
A time-series forecasting library used to model trends and seasonality in price data. It is well-suited for handling missing values and seasonal patterns.

Requests Library
Used to fetch real-time data from APIs with proper error handling.

Jupyter Notebook / Google Colab
Provides an interactive environment for development, testing, and experimentation.

CSV-based Storage
All datasets are stored in CSV format, making them easy to use and integrate.

Operating Environment
The system is designed to run on standard computers without

requiring high-end hardware, making it cost-effective and scalable.

VI. Results and Discussion

The proposed Nutritional Price Shock Index (NPSI) system was evaluated using both historical agricultural price data and real-time mandi data collected through the government API. The system focuses on protein-rich commodities and estimates protein affordability using a combination of time-series forecasting and real-time analysis.

A. Experimental Results

The system successfully processed multiple commodities and identified high-risk items based on their NPSI values. A sample of the computed results is presented in Table I.

TABLE I
Sample NPSI Results for Selected Commodities

Commodity	Expected Price	Actual Price	NPSI	Status
Arhar	0.032	0.051	1.59	Critical
Moong	0.028	0.041	1.46	Warning
Wheat	0.015	0.017	1.13	Normal
Soybean	0.010	0.012	1.20	Warning

The results indicate that commodities such as arhar and moong have relatively high NPSI values, suggesting a noticeable increase in the cost of protein compared to expected price trends.

B. Graphical Analysis

The system generates time-series plots that compare actual protein prices with forecasted values obtained from the Prophet model. These visualizations clearly highlight deviations from normal seasonal behavior.

The forecasting model effectively captures long-term trends and seasonal patterns, making it easier to identify abnormal spikes. The visible gap between actual and predicted values helps in understanding real-time deviations in a simple and intuitive way.

C. Performance Metrics

Although the primary objective of the system is shock detection rather than classification, the performance of the forecasting model can still be evaluated using general criteria:

- **Forecast Accuracy:** The Prophet model provides stable and reliable trend estimation with low error for seasonal data.
- **Execution Time:** The system processes multiple commodities within a short time, making it suitable for real-time applications.
- **Detection Efficiency:** The NPSI metric effectively identifies abnormal price increases across different commodities.

Unlike traditional classification systems, this approach does not rely on metrics such as precision or recall, as it is based on ratio-driven anomaly detection rather than supervised learning.

D. Comparison with Existing Methods

Traditional systems such as the :contentReference[oaicite:0]index=0 (CPI) and the :contentReference[oaicite:1]index=1 Food Price Index mainly track overall price inflation and commodity trends. Machine learning models like LSTM and Random Forest improve prediction accuracy, but they still focus only on raw price values.

In contrast, the proposed NPSI system offers several advantages:

- Evaluates affordability based on protein cost instead of raw prices
- Detects early warning signals of nutritional price shocks
- Integrates real-time data with forecasting techniques
- Provides clear and interpretable outputs for policy decision-making

TABLE II
Comparison with Existing Approaches

Method	Focus	Real-Time	Nutritional Analysis
CPI	Price Inflation	No	No
ML Models (LSTM, RF)	Price Prediction	Limited	No
FAO Index	Global Prices	No	No
Proposed NPSI	Nutrient Cost	Yes	Yes

E. Discussion

The results show that the proposed system provides a more meaningful way to understand food affordability compared to traditional price-based indices. By focusing on protein cost, it better reflects the nutritional impact of price changes.

The use of time-series forecasting allows the system to distinguish between normal seasonal variations and unusual price spikes. This is especially important for identifying early signs of potential food insecurity.

In addition, the system's ability to incorporate real-time data along with fallback mechanisms improves its reliability in practical scenarios.

Overall, the NPSI system offers an effective and practical approach for monitoring nutritional affordability and supporting informed policy decisions aimed at improving food security.

VII. Applications

The proposed Nutritional Price Shock Index (NPSI) system has several real-world applications in the domains of food security, public health, and economic policy. By focusing on nutrient affordability rather than raw commodity prices, the system provides meaningful insights that can support decision-making at multiple levels.

Food Security Monitoring

The NPSI system can be used by government agencies and international organizations to monitor food affordability in real time. By identifying sudden increases in the cost of protein-rich commodities, the system enables early detection of potential food insecurity and malnutrition risks.

Policy Planning and Intervention
Policymakers can utilize NPSI outputs to design targeted interventions such as releasing buffer stocks, implementing price controls, or providing subsidies. The classification of commodities into normal, warning, and critical categories helps prioritize resource allocation.

Public Distribution System Optimization
The system can support the optimization of public distribution systems by identifying high-risk commodities and regions. This allows authorities to ensure the availability of affordable protein sources to vulnerable populations.

Agricultural Market Analysis
Farmers, traders, and market analysts can use the system to understand price trends and anticipate market fluctuations. This can assist in making informed decisions regarding crop production, storage, and distribution.

Nutritional Health Monitoring
Healthcare organizations and researchers can use NPSI as an indicator of nutritional risk. By tracking the affordability of essential nutrients, the system can help identify populations at risk of protein deficiency.

Integration with Smart Systems
The proposed system can be integrated with smart dashboards, mobile applications, and IoT-based monitoring systems to provide real-time alerts and visualizations. This enhances accessibility and enables widespread adoption.

Overall, the NPSI system serves as a versatile tool that bridges the gap between economic price monitoring and nutritional analysis, making it highly valuable for real-world applications.

VIII. Advantages of the Proposed System

The proposed Nutritional Price Shock Index (NPSI) system offers several advantages over traditional food price monitoring approaches and existing machine learning models.

Nutrient-Based Analysis
Unlike conventional systems that focus only on raw commodity prices, the proposed system evaluates affordability based on protein cost. This provides a more meaningful and health-oriented measure of food accessibility.

Early Shock Detection
The integration of time-series forecasting enables the system to distinguish between normal seasonal variations and abnormal price spikes. This allows early detection of nutritional price shocks before they impact public health.

Real-Time Data Integration
The system incorporates live mandi data through government APIs, enabling real-time monitoring and up-to-date analysis of food prices.

High Interpretability
The NPSI metric is simple and easy to interpret, making it accessible to policymakers and stakeholders without requiring deep technical expertise.

Scalability
The modular architecture allows the system to be extended to additional commodities, nutrients, or regions with minimal modifications.

Cost-Effective Implementation
The system is implemented using open-source tools such as Python and Prophet, making it economically feasible and accessible for large-scale deployment.

Robustness and Reliability
The inclusion of fallback mechanisms ensures continuous operation even when live data sources are unavailable.

Efficient Processing
The system processes multiple commodities quickly with low computational overhead, making it suitable for real-time applications.

IX. Limitations

Despite its advantages, the proposed system has certain limitations that need to be addressed in future work.

Limited Nutrient Scope
The current implementation focuses only on protein as a measure of nutritional value. Other essential nutrients such as vitamins and minerals are not considered.

Dependence on Data Quality
The accuracy of the system depends on the quality and completeness of historical and real-time datasets. Missing or inconsistent data can affect prediction performance.

API Dependency
The system relies on external APIs for live data. Any failure or delay in the API may impact real-time analysis, although fallback mechanisms partially mitigate this issue.

Lack of External Factors
The forecasting model does not incorporate external variables such as weather conditions, government policies, or global market trends, which can influence price fluctuations.

Limited Testing Environment
The system has been tested on a specific dataset and geographical context. Its performance in other regions or markets may vary.

Model Limitations
While Prophet is effective for capturing seasonality, it may struggle with sudden and irregular price spikes, affecting prediction accuracy in highly volatile scenarios.

Computational Constraints for Scaling
Although efficient for moderate datasets, scaling the system to a very large number of commodities or regions may require additional optimization.

Simplified Protein Mapping
The system uses fixed protein values for commodities, which may not account for variations due to processing, quality, or regional differences.

X. Future Work

Although the proposed Nutritional Price Shock Index (NPSI) system demonstrates effective performance in detecting nutritional price shocks, several improvements can be explored in future research.

Incorporation of Deep Learning Models

Future work can explore advanced deep learning techniques such as Long Short-Term Memory (LSTM) networks, Transformers, or hybrid models to improve forecasting accuracy, especially for handling sudden price spikes.

Inclusion of Multiple Nutrients

The current system focuses only on protein as a measure of nutritional value. Future extensions can include other essential nutrients such as calories, iron, and vitamins to develop a comprehensive nutritional affordability index.

Integration of External Factors

Incorporating external variables such as weather conditions, rainfall patterns, government policies, and global market trends can improve the robustness and predictive capability of the model.

Real-Time Deployment and Dashboard Development

The system can be deployed as a real-time web-based dashboard or mobile application, enabling policymakers and stakeholders to monitor nutritional affordability continuously.

Expansion to Larger Datasets and Regions

Future work can extend the system to cover larger datasets, multiple regions, and international markets to improve generalization and scalability.

Integration with IoT and Smart Systems

The system can be integrated with IoT-based agricultural monitoring systems to provide real-time data collection and automated decision-making.

Geospatial Analysis

Incorporating location-based analysis can help identify region-specific nutritional risks and enable targeted interventions.

Overall, these enhancements can significantly improve the accuracy, scalability, and practical applicability of the NPSI system.

XI. Conclusion

This paper addressed the critical problem of food affordability by focusing on the cost of essential nutrients rather than raw commodity prices. Traditional systems such as the Consumer Price Index fail to capture the nutritional impact of price fluctuations, leading to hidden malnutrition. To overcome this limitation, the study proposed the Nutritional Price Shock Index (NPSI), a novel approach that measures affordability based on protein cost.

The proposed system integrates historical and real-time agricultural price data and applies time-series forecasting using the Prophet model to establish expected price trends. By comparing actual and predicted protein prices, the system successfully identifies abnormal price shocks and classifies them into different alert levels. The results demonstrate that the NPSI provides meaningful insights into nutritional affordability and enables early detection of potential food insecurity.

Overall, the proposed approach offers a practical, scalable, and interpretable solution for monitoring food security. It has significant implications for policymakers, researchers, and stakeholders by supporting informed decision-making and timely interventions to improve public health outcomes.

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